

Solution: From Theorem 4 on page 37, if A is an $m \times n$ matrix, the columns of A span $\mathbb{R}^m$ if and only if for each b in $\mathbb{R}^m$, the equation $Ax=b$ has a solution. $\therefore$ The columns of A do not span $\mathbb{R}^m$ if and only if for some b in $\mathbb{R}^m$, the equation $Ax=b$ has no solution. By defn. of inconsistency, if the columns of A do not span $\mathbb{R}^m$, then the equation $Ax=b$ is inconsistent for some b in $\mathbb{R}^m$.

$\therefore$ The given statement is True $\square$

25) Note that $\begin{bmatrix} 4 & -3 & 1 \\ 5 & -2 & 5 \\ -6 & 2 & -3 \end{bmatrix} \begin{bmatrix} -3 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -7 \\ -3 \\ 10 \end{bmatrix}$. Use this fact (and no row operations)

to find scalars c_1, c_2, c_3 such that $\begin{bmatrix} -7 \\ -3 \\ 10 \end{bmatrix} = c_1 \begin{bmatrix} 4 \\ 5 \\ -6 \end{bmatrix} + c_2 \begin{bmatrix} -3 \\ -2 \\ 2 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 5 \\ -3 \end{bmatrix}$

Solution: ~~From the data of $Ax=b$, we~~ From Theorem 3, we know that the matrix equation $Ax=b$ has the same solution set as the vector equation $x_1 a_1 + x_2 a_2 + \dots + x_n a_n = b$. $\therefore$ It's easy to see that $x_1 = c_1 = -3, x_2 = c_2 = -1$ and $c_3 = x_3 = 2$. $\square$

28) Rewrite the (numerical) matrix equation below in symbolic form as a vector equation, using symbols $v_1, v_2, \dots$ for the vectors and $c_1, c_2, \dots$ for scalars. Define what each symbol represents, using the data given in the matrix eqn.

$$\begin{bmatrix} -3 & 5 & -4 & 9 & 7 \\ 5 & 8 & 1 & -2 & -4 \end{bmatrix} \begin{bmatrix} -3 \\ 2 \\ 4 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ -1 \end{bmatrix}$$

Soln: From Theorem 3, the above matrix equation can be rewritten as the vector equation $c_1 v_1 + c_2 v_2 + c_3 v_3 + c_4 v_4 + c_5 v_5 = b$, where $c_1 = -3, c_2 = 2, c_3 = 4, c_4 = -1$ and $c_5 = 2$ are the scalars and $v_1 = \begin{bmatrix} -3 \\ 5 \end{bmatrix}, v_2 = \begin{bmatrix} 5 \\ 8 \end{bmatrix}, v_3 = \begin{bmatrix} -4 \\ 1 \end{bmatrix}, v_4 = \begin{bmatrix} 9 \\ -2 \end{bmatrix}, v_5 = \begin{bmatrix} 7 \\ -4 \end{bmatrix}$ and $b = \begin{bmatrix} 8 \\ -1 \end{bmatrix}$ are the vectors.